

Duke University Response to Assist Development of the Artificial Intelligence Action Plan

This is a response to the request for public comments to assist development of the Artificial Intelligence Action Plan, as a response to Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence) whose goal is to "establish U.S. policy for sustaining and enhancing America's AI dominance in order to promote human flourishing, economic competitiveness, and national security." Our responses consider that "The Trump Administration is committed to ensuring the United States is the undeniable leader in AI technology." We specifically address the role that U.S. universities can play in promoting human flourishing, economic competitiveness, and national security through actions by the Administration. We address topics of model development, open source development, applications, explainability and assurance of model outputs, cybersecurity, data privacy, technical standards, national security, research and development, education and workforce, innovation and competition, and energy needs. We will suggest concrete AI policy actions needed to address the topics raised, as encouraged in the Request for Information.

This response comes from Duke University. Duke and other top U.S. universities hold a unique role in enshrining the position and reputation of the United States as the global leader of AI technology. Duke trains top students entering the AI workforce, and its AI faculty collaborates across the spectrum of companies, non-profits, local and federal governments, and beyond on some of the most innovative AI technologies that exist today. Companies founded by and/or collaborating with its faculty are on the cutting edge of AI algorithm development and its applications across science, engineering, healthcare, and beyond. Top AI scientists and students (and top scientists and students in other areas, too) want to come to Duke because of its strong AI program. As is true in all such top universities, much of its most novel, risky, and important research is largely funded by federal grants.

The lead writer of this comment is a leader in interpretable AI, which is a more powerful version of explainable AI where algorithms are transparent – and not black box models. This is an extremely important area for trustworthy AI because it allows humans to double-check and troubleshoot the AI without needing to blindly trust it. This lead writer is a distinguished professor at Duke who works with neurologists, radiologists, otolaryngologists, heart monitoring specialists, materials scientists, reinforcement learning specialists, computational biologists, law professors, superior court judges, data visualization experts, computer vision experts, and musicians. Other writers of this comment are materials scientists, biomedical engineers, experts in natural language

processing on medical data, computer security, the efficiency of neural network training, computational neuroscience, medicine, and robotics, among others.

In the list below, we pose some important criteria for the U.S. to maintain its leadership role in AI, with suggested concrete AI policy actions.

- **We must have the best AI workforce.** U.S. universities such as Duke are currently the leading suppliers of employees to the world's top AI companies. Companies that currently lead the world in industrial AI technology hire mainly U.S.-trained students. For instance, Google, X, Meta, Amazon, and OpenAI extensively hire from U.S. universities including Duke. Currently, U.S. universities are generally viewed as the most desirable places for top talent to be trained and hired. If U.S. universities fail, that top talent (both students and faculty) will go to other countries. We must not let that happen. The reason this top talent views U.S. universities so highly is in large part because of the high quality of research and the caliber of faculty at these universities. Thus, as discussed later, the U.S. must unquestionably maintain the top university faculty in the world in AI. *We suggest that the AI Policy Action plan includes continued funding for research at U.S. universities to maintain the world's highest quality research, top faculty, and top students, as well as training programs to help universities train and recruit top students in AI.*
- **Our workforce must be trained using our U.S.-centered values:** If the world's top AI students were no longer trained within the U.S., we would have ceded control over how they are trained and the values they are trained on to other countries. This includes countries that have very strong AI programs, but currently cannot compete with the U.S. for students. Students trained elsewhere will not necessarily share a US-centered view of how to use their skill set to encourage human flourishing. We already know that other countries and other governments do not share the U.S. view of human flourishing, often using AI against their own citizens to stifle their freedoms. This is why U.S. universities should maintain strong control over their ability to train AI students in consideration of U.S. values. *Thus, we suggest that the AI Policy Action plan includes continued funding for training programs and research at U.S. universities on topics that embed our values into AI, including programs focused on trustworthy and ethical AI, control over AI systems, and applications of AI systems that involve U.S.-centered values. This includes work on interpretable AI (designing models that are self-explaining, as opposed to "explainable AI" where the goal is to explain a black box), the design of AI systems that are substantially easier to troubleshoot and use (not large and "clunky" black box systems), the design of large language models that are value-*

aware and use interpretable reasoning processes (as much as possible) so they are easier to troubleshoot, and how such systems should collaborate effectively with humans and improve the health and well-being of all people. It also includes applications, where universities work with nonprofits, charities, local government services, federal agencies, medical centers, scientists, legal experts, and other entities whose work we value, and that could be made much more efficient, but that could not hire AI specialists on their own. It also includes projects that allow individuals to study AI systems for possible security, medical, legal, and other risks.

- **The U.S. must have the top AI university faculty.** Currently, U.S. universities have the world's top AI faculty. This is why leading AI companies often hire ("poach") faculty from U.S. universities and why high-level executives of some leading AI companies are still university professors. Companies often hire faculty whose research, often having been funded by the federal government for many years through NSF/NIH/DOE/DOD/ etc. grants, is directly useful for their product development. Many companies arise directly from US universities and license technology from U.S. universities or spin-off companies, injecting innovation into and providing a boost to the U.S. economy. Many companies also have programs where faculty work part-time (e.g., one day per week) at a company, or consult with AI companies. If U.S. university jobs become less desirable due to a lack of governmental support, faculty at U.S. universities will leave, and the U.S. AI industry and the related economy will be severely weakened. Thus, *we again suggest that the AI Policy Action plan includes continued funding for research at U.S. universities to maintain the world's highest quality research, top faculty, and top students.* Faculty need access to resourced computing facilities for training AI models. Thus, *we recommend that the AI Policy Action Plan include funding for computing systems at U.S. universities.* Currently, many of these computing systems are forced into overhead (i.e., indirect costs), which are currently under consideration for major reductions, making it unlikely that many departments will be able to maintain their computing clusters and staff, stifling research and U.S. competitiveness.
- **U.S. universities must continue to produce and maintain public AI software that is widely used and freely available.** This software is often used by or adapted for use in U.S. companies, as well as U.S. governmental and other organizations. For instance, publicly available software created by the lead author of this response is used by the NYPD, Con Edison (the power company for New York City), in intensive care units at hospitals across the country, by numerous bioinformatics experts, and by anyone else who wishes to use it. This software

makes our power systems more resilient to threats, makes our streets safer from crime, makes our medical care better, and improves our ability to make scientific discoveries. University-designed software is typically produced in collaboration with students who are funded by the federal government. If U.S. universities are weakened, either through unavailability of grant funding for students, research, computational and cybersecurity infrastructure, facilities, etc., then organizations throughout the U.S. will be less efficient, since AI software will not be as readily available to them or will fall into disrepair due to lack of maintenance. *We recommend that the AI Policy Action plan includes support for researchers and the resources they need to develop and maintain publicly-available software.*

- **Universities are (and have always been, and must continue to be) the key to filling the gap between what is *needed* for human flourishing and what profit-making entities *want* to create.** This will be particularly true if the U.S. is aiming to maintain its stance as the world leader in AI. For instance, U.S. universities often collaborate with U.S. organizations that cannot afford to develop their own AI tools. Because U.S. universities have projects focused on human flourishing, they often collaborate with charities, medical centers, legal experts and local governments to create, implement, and maintain AI technology that is customized for their uses. This would be unaffordable and out-of-reach for those entities otherwise. Given how valuable we know AI can be, with the proper safeguards, universities have a key role to play in human flourishing and the efficiency of public systems. Importantly, ***universities can aim directly for goals related to human flourishing without the entanglement and distraction of profits.*** As shown in the examples above and below, ***profit for AI companies is not necessarily or even generally aligned with human flourishing.***
 - An example described in The Verge ("What happens when an algorithm cuts your health care" <https://www.theverge.com/2018/3/21/17144260/healthcare-medicaid-algorithm-arkansas-cerebral-palsy>) illustrates what happens when proprietary black box AI models do not allocate care correctly for individuals with severe chronic illnesses in Arkansas. Here, the decisions were based on accidentally using the *wrong AI algorithm*, which is shocking. The company didn't realize the mistake until they were sued and were forced to look at the algorithm's details. A more responsive and responsible collaborator would have easily figured that out. Universities can easily create such algorithms more cost-effectively for the state, make these algorithms more transparent, and be better collaborators.
 - Another example is AI "add-ons" in IVF treatment. The IVF industry "is notorious for aggressively marketing unproven clinical and laboratory 'add

ons" ("Ethical Implementation of Artificial Intelligence to Select Embryos in In Vitro Fertilization " Afnan et al., 2021, <https://arxiv.org/pdf/2105.00060>), where IVF clinics charge patients thousands of dollars to use AI tools to predict whether an embryo will lead to a fetal heartbeat pregnancy, based on images of it. Afnan et al., 2021, describe a situation from two earlier works, where the first paper claimed almost perfect performance for such an AI tool, and the second paper pointed out that the first paper had artificially inflated its reported performance by changing its dataset to include easy cases that IVF clinics would find obvious. This directly illustrates the conflict between business drivers and human flourishing in a literal way – AI tool manufacturers were incentivized to produce almost perfect performance even if it involved choosing the wrong embryos for IVF that would never implant – and they were potentially charging patients thousands of dollars to make these predictions. Universities can help detect and mitigate these situations. They can also create alternative models that are publicly available that explain their decisions to doctors in a way they could evaluate transparently.

- As a third example, an author of this document is an author of the only AI algorithm currently widely used in critical care brain monitoring. The algorithm is in the form of a simple scorecard. It is extremely small, fits on an index card, and can be memorized by a doctor. It is also as accurate as the best deep neural network for the data it was trained on. This is an example of a technology that doesn't force doctors to trust it – doctors can evaluate this technology for themselves since they are not black box models, are not proprietary, are easy to understand, and are freely available. This algorithm saves lives, and its development was funded by the federal government. A company working on this problem could not have achieved this level of transparency and still have made a profit from it.

Thus, we recommend that the AI Policy Action plan includes research support that works with entities whose main goal is some type of human flourishing.

- **Keeping the NIH and NSF strong will allow the U.S. to continue to make leaps in human health technologies.** The lab of the lead author of this document recently discovered a new biomarker that can predict breast cancer up to 5 years in advance, published in the top journal *Radiology*. This work was funded by the NSF and could save millions of people through early detection, and could save millions of dollars in making breast cancer screening more efficient. Two other authors have used AI to develop noninvasive methods to detect diabetes, work funded by the NIH. While the general topic of AI and health could be discussed at length, to keep this document short, we will not list the numerous ways that AI

contributes to health technologies. AI in medical care would continue to allow the US to develop new drug targets, categorize and treat chronic illnesses, detect diseases much earlier than before at a level of accuracy not seen before, find ways to empower individuals who can no longer communicate or act normally due to a medical condition, make medical care more widely available through AI-tools, and make medical care much more efficient with AI-human collaborations. *Thus, we recommend that the AI Policy Action Plan fully support NIH and NSF for healthcare-related and biomedical AI.*

- **Keeping the NSF strong will allow the U.S. to keep its lead in science and technology, which are currently being revolutionized by AI.** U.S. universities are capable of solving a multitude of scientific and engineering problems that companies cannot. Universities regularly engage in research that is too high risk for companies to engage in. If the development of AI technologies is relegated to the few companies that can afford to employ individuals to develop high-risk early-stage technologies, innovation will be highly stifled. This is particularly important right now, as AI technologies are changing extremely quickly and we need a nimble and intelligent workforce just to keep up with its capabilities and possibilities. AI is currently rapidly advancing areas of materials science, biology, chemistry, physics, astronomy, zoology, neuroscience, etc. Funding to universities from the NSF is the key to keeping the U.S. in the lead in these fields. *Thus, we recommend that the AI Policy Action plan include funding for scientific discovery at universities.*
- **Universities serve as a unique hub for AI + X research and technology innovation.** Universities like Duke provide an indispensable environment for integrating AI with a wide array of other disciplines (“X”), such as materials science, social science, robotics and autonomy, engineering design, healthcare, and edge computing. The academic setting fosters open collaboration and cross-pollination of ideas in a way that industry alone cannot replicate. This open, interdisciplinary culture makes universities a force multiplier: innovations in “X” catalyze advances in AI, which in turn empowers further breakthroughs in “X.” Duke’s breadth of strengths across these domains positions it and similar top U.S. universities as the ideal crucible for pioneering AI + X technology. The discoveries and insights gained here can then be shared widely, free from many of the profit-driven constraints that limit purely industry-led endeavors, ultimately delivering social, technological, and economic benefits that also feed back into advancing AI itself. *Thus, we recommend that the AI Policy Action Plan provide strong funding for interdisciplinary AI + X research in university environments.*

- **Universities are likely to be the key to dealing with the multitude of AI scams, particularly coming from foreign countries, that are starting to arise online and across phone lines.** Modern Generative AI is exceedingly convincing. So far, *no entity* (no company, no government) has been able to stop these scammers within the U.S. While we can construct fences around our homes to keep criminals physically out, we cannot keep them out of our children's devices – in direct communication with our children, often without our knowledge. We cannot keep them from our older relatives. We cannot keep scammers out of our phones either; scammers sometimes even use our own family members' biometrics to convince us to send money to fraudsters. Novel technologies developed at U.S. universities can help combat these problems. Surely the U.S. cannot righteously claim that it is the world leader in AI if billions of dollars are regularly stolen from U.S. consumers using AI technology from other countries. Scammers using AI pose national security issues. *We recommend that the AI Policy Action plan take a multi-pronged attack, involving academic researchers equipped with novel and powerful AI, particularly those in computer security, companies, government agencies empowered by new technologies, legal experts, and non-profits like C2PA that encourage the use of digital signatures for establishing provenance of content.*
- **Universities are essential for developing critical AI for national security.** It is a reality that AI can pose serious challenges for U.S. national security and we must use the intellectual strength of our academic environment to develop technologies that mitigate or counter threats. The war in Ukraine for example, has hypercharged the development of AI-based defense technology – this is a serious national security concern for the U.S. Also, other countries are rapidly developing their AI capabilities to affect (and weaken) democracies, thereby extending their regional influence. We work collaboratively with government research labs to develop promising groundbreaking new technologies, and ensure they are put to use for the U.S. *Thus, we recommend that robust support of AI development for national security purposes be included in the AI Policy Action plan.*
- **Encouragement of economic competitiveness comes with reducing monopolistic practice.** By definition, monopolies stifle competitiveness. The Administration can remove barriers to innovative research and innovative products by continuing to reduce AI Tech monopolistic practices. This would allow more choices for consumers and more opportunities for academics to engage with a variety of industry participants. Academics could then engage with companies much more broadly – not just in the ways that current Big Tech companies have limited academic researchers, where data is often difficult to access, and studies are impossible to conduct because of lack of access. There are many research

questions that academic researchers could potentially tackle (for instance, to study the ways that technologies affect individuals, and to develop AI tools that many individuals would enjoy), but Big Tech monopolistic practices hinder them from engaging in these studies because of these companies' concerns of how it might affect their reputation and their profits and create potential disruptions to their business model. Access to data is key. Competition among companies would make this possible. *Thus, we recommend that AI Policy Action include a plan to reduce monopolistic practices.*

- **Encouragement of economic competitiveness reduces risks to consumers due to database hacking.** Some large companies have amassed huge amounts of highly personal data about millions of U.S. consumers, often including sensitive biometrics, that no other company or university can collect and use. This creates a huge economic advantage for these companies. Monopolistic practices allow them to collect the data, and they cement their advantage by refusing to allow others to collect it or use it. These data collection practices also increase risks to the public. AI software can turn that data into very digestible and very dangerous forms of information, including facial recognition from their biometrics, determination of someone's individual vulnerability to fraud based on their browsing history, extraction of someone's personal medical history (e.g., from just their search queries and YouTube history), and even fundamentally what that person is thinking about (e.g., browser history, email, photo stream, advertisement views and responses, chat room conversations, and even simultaneous wearable devices data). Hackers steal data from large organizations all the time. Human flourishing includes keeping the public's personal data secure; the U.S. cannot possibly be the world leader in AI if it allows millions of individuals' personal data to be stolen regularly. *Thus, we recommend that AI Policy Action includes data security at such companies to match the new risks from AI, by taking measures to reduce monopolistic practice, which has the secondary effect of reducing databases where data is highly concentrated.*
- **Energy availability is critical for strong AI capabilities.** The rise of AI comes with increasing energy demand, including demand fluctuations that can place significant strain on energy grids across large regions. There are at least three aspects by which sufficient energy availability for current and future AI needs can be ensured, including (1) reducing energy use for AI outcomes by algorithmic/software optimization, (2) more energy-efficient AI hardware, (3) developing and provisioning economical energy sources of sufficient power and availability. Researching energy-efficient AI, energy sources and resilient power grids is, therefore, a key collateral requirement for deploying a resilient, leading AI

infrastructure in the U.S. *Thus, we recommend that AI development be flanked by funding research into all types of energy technologies that can guarantee a sufficient, secure supply of energy for robust growth of current and future AI technologies.*

We hope this comment is useful to the Administration in setting its priorities on maintaining US AI dominance.

Sincerely,

Cynthia Rudin

Distinguished Professor of Computer Science, Electrical and Computer Engineering, Statistics, Biostatistics & Bioinformatics, and Mathematics

Ehsan Abadi

Associate Professor of Radiology and Electrical and Computer Engineering

Monica Agrawal

Assistant Professor of Biostatistics & Bioinformatics, Computer Science, and Biomedical Engineering

David Banks

Professor of the Practice of Statistics

Alberto Bartesaghi

Associate Professor of Computer Science, Biochemistry and Electrical and Computer Engineering

Volker Blum

Associate Dean for Research, Pratt School of Engineering and Timothy P. and Mary M. Rooney Associate Professor of Mechanical Engineering and Materials Science and of Chemistry

Brinnae Bent

Executive in Residence in the Engineering Graduate and Professional Programs

L. Cate Brinson

Distinguished Professor of Mechanical Engineering and Materials Science

David Carlson

Associate Professor of Civil and Environmental Engineering

Boyuan Chen

Assistant Professor of Mechanical Engineering and Materials Science

Tingjun Chen

Assistant Professor of Electrical and Computer Engineering and Computer Science

Yiran Chen

Distinguished Professor of Electrical and Computer Engineering

Leslie Collins

Professor of Electrical and Computer Engineering

Bhuwan Dhingra

Assistant Professor of Computer Science

Bruce Donald

Distinguished Professor of Computer Science, Mathematics, and Chemistry

Jessilyn Dunn

Assistant Professor of Biomedical Engineering and Biostatistics & Bioinformatics

Pardis Emami-Naeini

Assistant Professor of Computer Science, Electrical and Computer Engineering, and Public Policy

Ethan Fang

Associate Professor of Biostatistics & Bioinformatics and Electrical and Computer Engineering

Aaron Franklin

Associate Dean for Faculty Affairs, Professor of Electrical and Computer Engineering

Johann Guilleminot

Associate Professor of Mechanical Engineering and Materials Science

Ricardo Henao

Associate Professor of Biostatistics & Bioinformatics

Simon Mak

Assistant Professor of Statistical Science

Kamesh Munagala

Professor of Computer Science

Willie Padilla

Distinguished Professor of Electrical and Computer Engineering

David Page

Distinguished Professor and Department Chair of Biostatistics & Bioinformatics,
Professor of Computer Science

Ron Parr

Professor of Computer Science

Jian Pei

Distinguished Professor and Department Chair of Computer Science, Electrical and
Computer Engineering, and Biostatistics & Bioinformatics

Tania Roy

Assistant Professor of Electrical and Computer Engineering

Ehsan Samei

Distinguished Professor of Radiology

Rohit Singh

Assistant Professor of Biostatistics & Bioinformatics, Cell Biology, Computer Science, and
Electrical and Computer Engineering

Stacy Tantum

Associate Professor of the Practice of Electrical and Computer Engineering

Carlo Tomasi

Distinguished Professor of Computer Science

Yu Tong

Assistant Professor of Mathematics

Yesenia Velasco

Lecturer of Computer Science

Emily Wenger

Assistant Professor of Electrical and Computer Engineering

Pan Xu

Assistant Professor of Biostatistics & Bioinformatics, Computer Science, and Electrical & Computer Engineering

Xiaowei Yang

Professor of Computer Science

Anru Zhang

Associate Chair for Research and Associate Professor, Department of Biostatistics & Bioinformatics

Danyang Zhuo

Assistant Professor of Computer Science

Submitted for the institution by:

Jenny Lodge,

Vice President for Research & Innovation

Duke University, Durham, NC

March 15, 2025

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.